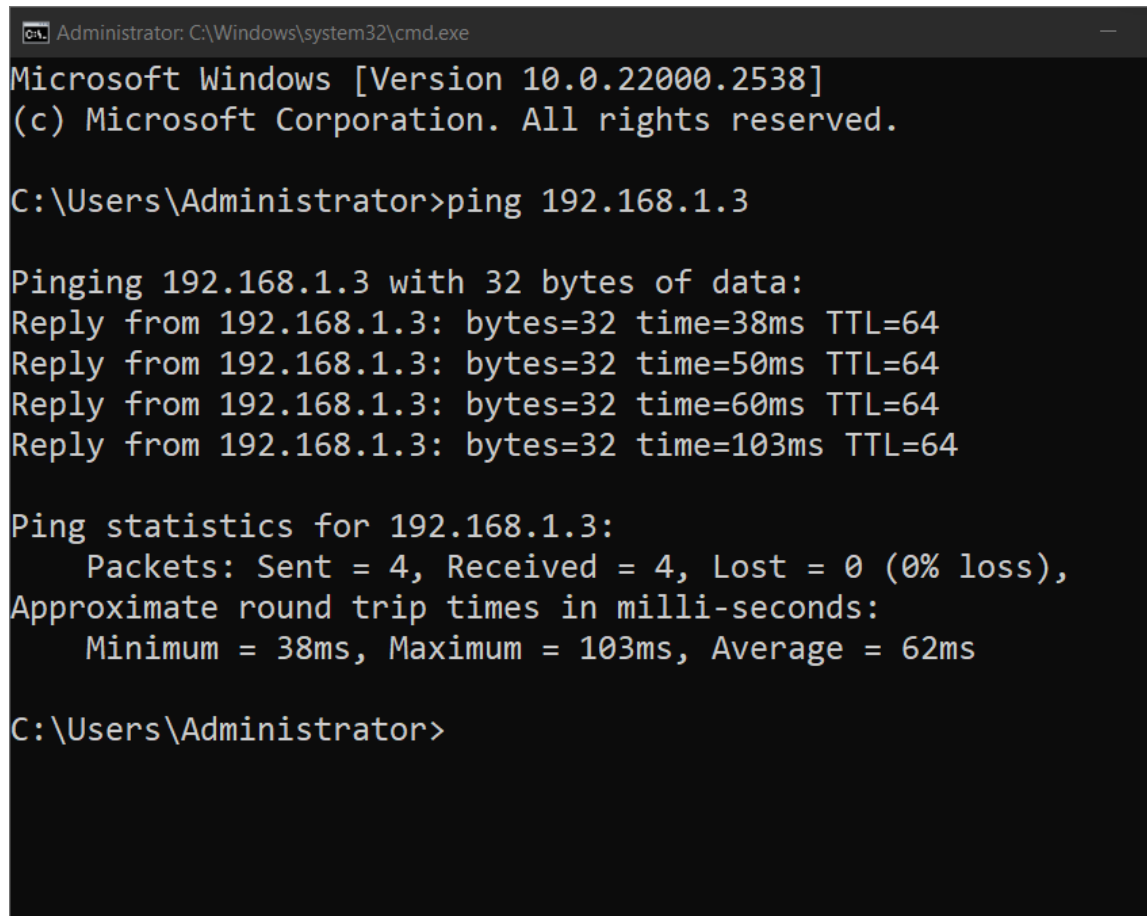


Networking testing practical

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.

Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

A screenshot of a Windows Command Prompt window. The title bar reads "Administrator: C:\Windows\system32\cmd.exe". The window content shows the Windows version "10.0.22000.2538" and copyright information. The user has entered the command "ping 192.168.1.3". The output shows four successful replies from 192.168.1.3 with varying response times (38ms, 50ms, 60ms, 103ms) and a TTL of 64. Ping statistics are also displayed, showing 4 packets sent, 4 received, 0 lost, and an average round trip time of 62ms.

```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:
Reply from 192.168.1.3: bytes=32 time=38ms TTL=64
Reply from 192.168.1.3: bytes=32 time=50ms TTL=64
Reply from 192.168.1.3: bytes=32 time=60ms TTL=64
Reply from 192.168.1.3: bytes=32 time=103ms TTL=64

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 103ms, Average = 62ms

C:\Users\Administrator>
```

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

➤ tracert www.youtube.com

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

= Request timed out

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

IP Address	Device/Gadget	Owner
192.168.1.1	Wi-Fi Router	Home Router
192.168.1.2	Laptop	Me
192.168.1.3	Smartphone	Me
192.168.1.4	Smart TV	Family
192.168.1.5	Friend's Phone	Friend

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.

Hint: Include screenshots or command outputs if possible.

➤ Network Setup Report

I connected my laptop and another device to the same Wi-Fi network. I checked the IP address using `ipconfig` and tested the connection with the `ping` command. The ping was successful, and after disconnecting one device, it showed Request timed out.

I also ran `tracert www.youtube.com` to see the route to YouTube, and it displayed multiple network hops. Overall, the network worked correctly, and the tests confirmed proper connectivity.